OBJECT ORIENTED PROGRAMMING (OOP)

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AGENDA

- INTRODUCTION TO SUPER KEYWORD
- USAGE OF SUPER KEYWORD
- EXAMPLES OF SUPER KEYWORD
- STATIC KEYWORD
- STATIC VARIABLE
- STATIC METHOD
- STATIC BLOCK
- INNER CLASSES

SUPER KEYWORD

- The **super** keyword in Java is a reference variable which is used to refer immediate parent class object.
- Whenever you create the instance of subclass, an instance of parent class is created implicitly which is referred by super reference variable.

USED OF SUPER KEYWORD

1. super is used to refer immediate parent class instance variable.
   - We can use super keyword to access the data member or field of parent class. It is used if parent class and child class have same fields.
2. super can be used to invoke parent class method
3. super is used to invoke parent class constructor.

```java
class Animal{
String color="white";
}
class Dog extends Animal{
String color="black";
void printColor(){
System.out.println(color);//prints color of Do
g class
System.out.println(super.color);//prints color
of Animal class
}
}
class TestSuper1{
public static void main(String args[]){
Dog d=new Dog();
d.printColor();
}}
```



```java
class Animal{
void eat(){System.out.println("eating...");}
}
class Dog extends Animal{
void eat(){System.out.println("eating bread...");}

void bark(){System.out.println("barking...");}
void work(){
super.eat();
bark();
}
}
class TestSuper2{
public static void main(String args[]){
Dog d=new Dog();
d.work();
}}
```

```java
class Animal{
Animal(){System.out.println("animal is ceated");}
}
class Dog extends Animal{
Dog(){
super();
System.out.println("dog is created");
}
}
class TestSuper3{
public static void main(String args[]){
Dog d=new Dog();
}}
```

REAL APPLICATION OF SUPER KEYWORD

```
class Person{
int id;
String name;
Person(int id,String name){
this.id=id;
this.name=name;
} }
class Emp extends Person{
float salary;
Emp(int id,String name,float salary){
super(id,name);//reusing parent constructor
this.salary=salary;
}
void display(){System.out.println(id+" "+name+" "+salary);}
}
class TestSuper5{
public static void main(String[] args){
Emp e1=new Emp(1,"ankit",45000f);
e1.display();  }}
```

STATIC KEYWORD

- The **static keyword** in Java is used for memory management mainly.
- We can apply static keyword with variables, methods, blocks and nested classes.
- The static keyword belongs to the class than an instance of the class.

STATIC KEYWORD CAN BE:

- The static can be:

1. Variable (also known as a class variable)
2. Method (also known as a class method)
3. Block
4. Nested class

I) JAVA STATIC VARIABLE

- The static variable can be used to refer to the common property of all objects (which is not unique for each object), for example, the company name of employees, college name of students, etc.
- The static variable gets memory only once in the class area at the time of class loading.
- Example of Java static variable using counter variable

JAVA STATIC METHOD

- If you apply static keyword with any method, it is known as static method.
- A static method belongs to the class rather than the object of a class.
- A static method can be invoked without the need for creating an instance of a class.
- A static method can access static data member and can change the value of it.

EXAMPLE OF STATIC METHOD

```
class Student{
    int rollno;
    String name;
    static String college = "ITS";
    static void change(){
    college = "BBDIT";
    }
    Student(int r, String n){
    rollno = r;
    name = n;
    }
    void display(){System.out.println(rollno+" "+name+" "+coll
ege);}
}
```

```
public class TestStaticMethod{
    public static void main(String args[]){
    Student.change();
    Student s1 = new Student(111,"Karan");
    Student s2 = new Student(222,"Aryan");
    Student s3 = new Student(333,"Sonoo");
    //calling display method
    s1.display();
    s2.display();
    s3.display();
    }
}
```

TWO MAIN RESTRICTIONS OF STATIC METHOD

1. The static method can not use non static data member or call non-static method directly.
2. this and super cannot be used in static context.

Q) WHY IS THE JAVA MAIN METHOD STATIC?

- Ans) It is because the object is not required to call a static method. If it were a non-static method, JVM creates an object first then call main() method that will lead the problem of extra memory allocation.

Q) IS THERE ANY VARIABLE, METHOD, FUNCTION WHICH CAN EXECUTED BEFORE MAIN METHOD?

- Answer is YES! It is static block
- Is used to initialize the static data member.
- It is executed before the main method at the time of classloading.

```
class A2{
  static{System.out.println("static block is invoked");}
 public static void main(String args[]){
  System.out.println("Hello main");
 }
}
```

INNER CLASSES

- A class inside another class is known as inner classes
- There are two types of inner classes
  - Member class
  - Static member class
- An inner class can:
  - Access the private members of the containing class
  - Have a private access modifiers

THANK YOU